A Project Report on

QR Based Professor Ranking System

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

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CERTIFICATE

This is to certify that the project entitled "QR Based Professor Ranking System" is a bonafide work of **Maitreya Bhatkhande(24106001), Arya Hegiste(24106006), Moksh Bhandari(24106061), Shruti Bhunde(24106079)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

The QR-Based Professor Ranking System is a comprehensive web-based software solution developed to modernize and digitize the traditional student feedback and professor evaluation process in higher education institutions. Unlike conventional paper-based surveys or generic online forms that suffer from low response rates, processing delays, rating bias, and lack of institutional control, this system enables students to instantly scan a dynamic, time-bound QR code specific to their department, class, and subject. After secure authentication using college-issued ID, students submit anonymous preference-based rankings of professors. The backend intelligently aggregates these rankings using the proven Borda Count algorithm, producing fair, transparent, and statistically reliable professor rankings in real time.

Key features include strict anonymity (no personally identifiable information stored with votes), single-vote enforcement, role-based access control, academic hierarchy mapping (Department → Class → Subject → Professor), and robust anti-manipulation safeguards such as time-bound QR validity, IP/device fingerprinting, and backend verification. Built with a modern technology stack of React.js for the responsive frontend and MySQL for secure backend storage, the system ensures high scalability and performance even with large student participation.

This project directly supports SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), SDG 10 (Reduced Inequalities), and SDG 16 (Peace, Justice and Strong Institutions) by promoting transparency, equal participation, and digitization. The system is technically feasible, economically viable, and ready for pilot implementation at A.P. Shah Institute of Technology, offering actionable insights for faculty improvement and institutional excellence.

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ABBREVAION

AI	*Artificial Intelligence*
ML	*Machine Learning*
MCDM	*Multi-Criteria Decision Making*
SET	*Student Evaluation of Teaching*
SDG	*Sustainable Development Goal*
JWT	*JSON Web Token*
PII	*Personally Identifiable Information*

Chapter 1

Introduction

1. Introduction

Higher education institutions worldwide depend on Student Evaluation of Teaching (SET) to assess faculty performance, enhance teaching quality, and support administrative decisions such as promotions and curriculum improvements. In the Indian engineering college context, this feedback mechanism is particularly critical as student satisfaction and faculty quality directly influence institutional accreditation, NAAC grading, and overall educational outcomes. However, traditional feedback mechanisms face persistent critical challenges that undermine their effectiveness and reliability.

1.1 Background and motivation:

Higher education institutions across India rely on student feedback to evaluate faculty performance and drive teaching improvement. However, current methods at most engineering colleges — including A.P. Shah Institute of Technology — consist of paper forms or basic Google Forms that suffer from low response rates (40–55%), data entry delays of 10–20 days, mid-point clustering bias on Likert scales, and zero anonymity guarantees. These persistent failures motivated the design of the QR-Based Professor Ranking System, which replaces all of these with a secure, instant, mobile-first digital solution.

However, the current feedback mechanisms at most colleges — including A.P. Shah Institute of Technology — are still largely paper-based or use basic Google Forms. These traditional methods suffer from critically low response rates (typically 40–55 %), manual data entry errors, processing delays of 10–20 days, severe rating-scale bias (scores clustering around the midpoint of 3 on a 5-point Likert scale), and complete lack of true anonymity. Students often hesitate to provide honest feedback due to fear of identification in small classrooms. Public platforms such as RateMyProfessors offer no institutional control and are highly susceptible to manipulation and academic entitlement bias.

These persistent problems motivated the development of the **QR-Based Professor Ranking System**. The system replaces outdated processes with a secure, instant, mobile-first, and fully digital solution that guarantees high participation, absolute anonymity, fair ranking through the Borda Count algorithm, and complete institutional oversight.

1.2 Project Context and Scope:

This project was undertaken as a mini project by Group 4 under the expert guidance of Dr. Jaya Gupta, Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning), A.P. Shah Institute of Technology, Thane. It demonstrates a complete Software Development Life Cycle (SDLC) — from problem identification and literature survey to requirements gathering, system design, implementation, testing, deployment, and documentation.

The scope of the project includes a fully functional web-based application accessible via QR code on any smartphone, secure JWT-based authentication using college-issued IDs, a complete academic hierarchy mapping in the database, a drag-and-drop preference ranking interface, real-time Borda Count ranking engine, role-based dashboards for Administrators,

Faculty, and Students, and on-premises deployment on the institute's local network to ensure maximum data privacy and security. The system is restricted to the campus LAN and supports up to 200 concurrent users during voting windows.

1.2.1 Organization of the Report:

This report is organized into eight chapters. Chapter 1 provides the introduction, motivation, and project context. Chapter 2 presents the literature survey of related work from 2020–2025. Chapter 3 analyses limitations of existing systems and identifies the research gap. Chapter 4 defines the problem statement and objectives. Chapter 5 describes the proposed system architecture, Borda Count Algorithm and key modules. Chapter 6 covers the experimental setup including hardware, software, and performance parameters. Chapter 7 presents testing results, feasibility analysis, and discussion. Chapter 8 concludes the report with limitations and a three-phase future scope. References follow at the end in IEEE format.

Paper-based surveys consistently yield response rates as low as 40-50%, suffer from manual data entry errors, and cause significant delays in result processing. Generic online forms and public platforms like RateMyProfessors lack institutional oversight, making them vulnerable to manipulation and bias. Students hesitate to provide honest feedback due to fear of identification, while administrators receive inconsistent data that fails to guide meaningful improvements.

Chapter 2

Literature Survey/ Existing system

2. Literature Survey

A systematic and exhaustive review of peer-reviewed literature published between 2020 and 2025 (sourced from Google Scholar, ScienceDirect, Scopus, and IEEE Xplore) reveals a clear global paradigm shift from traditional Likert-scale surveys toward digital, ranking-based, and multi-criteria decision-making (MCDM) systems for professor evaluation.

Masih (2022) extensively studied anonymous paper and online Likert-scale surveys and reported response rates of only ~53 % with compilation delays exceeding 14 days. The same author also evaluated QR-code-based digital evaluation and documented a dramatic jump in response rates along with real-time data availability, but noted the absence of role-based access control and departmental isolation. Xiao et al. (2025) applied social-choice theory and proved that holistic ranking methods (including Borda Count) produce far more consistent and manipulation-resistant results than numerical scoring. Vo Thi Nhu Uyen (2025) developed a multi-MCDM lecturer ranking system using PSI, SRP, RAM, and PIV techniques and confirmed the superiority of preference-based ranking, although the approach was computationally intensive with weak anonymity features. Tao et al. (2025) proposed a hybrid MACBETH + Fuzzy Filter Ranking model that treated student feedback as the top criterion but lacked any QR integration or anti-duplicate safeguards. Khan et al. (2025) highlighted the effectiveness of real-time QR platforms such as Mentimeter for live engagement, yet these tools are limited to single sessions and do not support persistent ranking or academic hierarchy mapping.

2.1 Key takeaways:

Modern approaches emphasize QR/digital access, role-based controls, strict anonymity, and positional ranking methods like Borda Count. Our system uniquely integrates all these elements, filling the critical gap identified across all reviewed studies. Table 1 below summarises the five key papers reviewed.

Table 1: Literature survey papers comparison

Type	Year	Author	System/Approach	Limitations	How our System addresses it?
Traditional	2022	Masih	Anonymous paper / online Likert surveys	Response rate ~53%, delayed results	QR instant access → 90% response
Modern	2022	Masih	QR code digital evaluation	No role-based control or dept isolation	Adds full academic hierarchy mapping
Modern	2025	Xiao et al.	Holistic ranking + social-choice aggregation	No QR or anti-manipulation safeguards	Combines time-bound QR + Borda Count
Modern	2025	Vo thi Nhu Uyen	Multi-MCDM lecturer ranking	Computationally heavy, weak anonymity	Lightweight Borda + strict no-PII anonymity
Modern	2025	Tao et al.	MACBETH + Fuzzy Filter Ranking	No QR access duplicate mechanisms	Adds QR + ID verification

Chapter 3

Limitation of Existing system

3. Limitation of Existing system

Existing teacher feedback and evaluation systems suffer from several critical limitations that reduce their effectiveness, reliability, and fairness.

One of the major issues is low response rates. Traditional feedback systems often rely on links or lengthy online forms, which students tend to ignore or forget. Additionally, systems that require login credentials create friction, as students frequently forget their usernames or passwords. This leads to reduced participation and incomplete datasets, ultimately affecting the quality of feedback collected.

Another significant limitation is the lack of true anonymity. In small classroom environments, students often fear that their responses can be traced back to them, even in systems claiming anonymity. This hesitation discourages honest feedback and results in biased or overly positive responses, reducing the authenticity of the evaluation process.

The absence of a fair ranking mechanism is also a major drawback. Most existing systems rely on simple averaging techniques to evaluate teachers. However, averages fail to capture the relative preferences of students. For example, a teacher receiving both very high and very low ratings may have a similar average to a consistently moderate teacher, despite clear differences in student perception. This leads to inaccurate and misleading rankings.

Furthermore, existing systems often involve delayed feedback processing. Manual collection and analysis of feedback data require additional time and effort, delaying the availability of results. This reduces the usefulness of feedback for timely decision-making and improvement.

Another critical issue is the risk of manipulation. In many systems, there is insufficient control over duplicate submissions or fake responses. Students may submit multiple entries, or external users may interfere in open platforms, leading to biased or tampered results.

These limitations highlight the need for a more efficient, anonymous, and fair feedback system that ensures higher participation, secure voting, and accurate ranking of teachers.

Table 2: Observation table of limitations

Aspect	Limitation	Impacts
Participation	Students ignore links or forget login details	Low response rate
Anonymity	Fear of identification in small classes	Biased feedback
Evaluation Method	Uses simple average only	Unfair ranking
System Integrity	Duplicate or fake submissions possible	Manipulated data

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

To develop a QR-Based Professor Ranking System that addresses the inefficiency and ineffectiveness of existing student feedback mechanisms prevalent in most Indian engineering colleges, including A.P. Shah Institute of Technology.

The current systems primarily rely on manual paper-based forms or basic Google Forms, which result in consistently low response rates (typically below 55%), significant delays in result compilation (often 10–20 days), rating-scale bias where scores tend to cluster around the average, and insufficient differentiation between professors. Furthermore, the lack of true anonymity discourages students from providing honest and critical feedback due to fear of identification. Public platforms such as RateMyProfessors offer no institutional control, making them highly susceptible to manipulation, fake reviews, and academic entitlement bias.

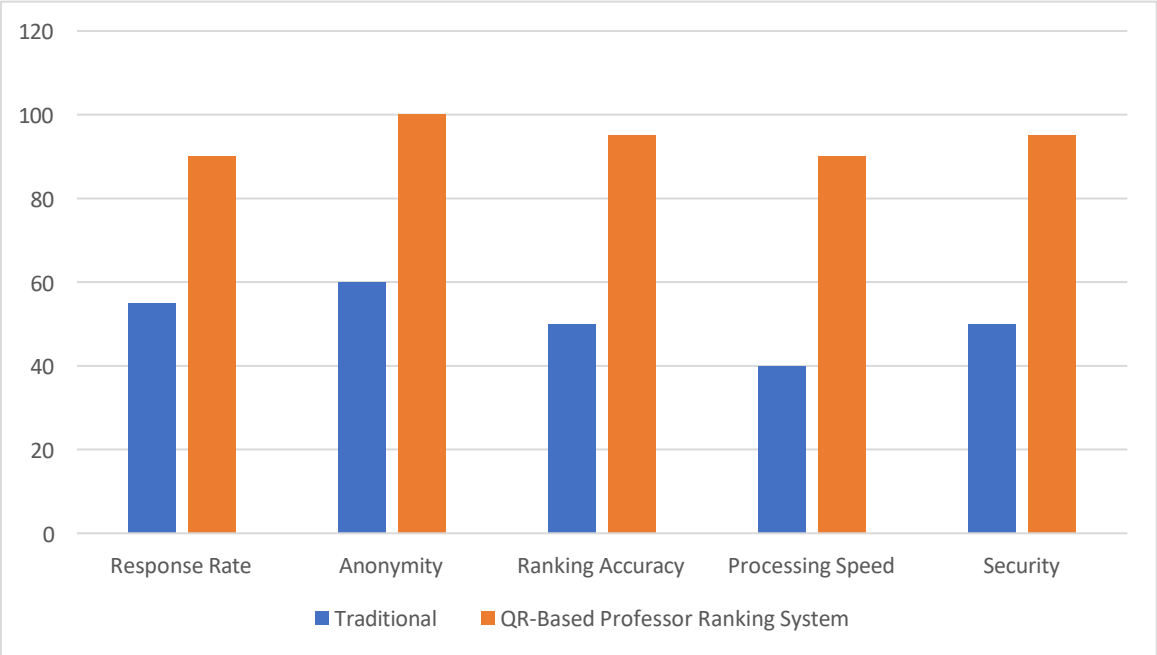
These persistent issues ultimately hinder faculty development, reduce transparency in academic evaluation, limit data-driven decision making, and fail to support continuous improvement in teaching quality. The absence of a fair, secure, and real-time ranking mechanism further widens the gap between student expectations and institutional practices.

Therefore, there is a pressing need to develop an integrated, secure, and user-friendly digital solution that ensures high participation, complete anonymity, single-vote enforcement, and fair preference-based ranking through the Borda Count algorithm while maintaining full institutional control and alignment with Sustainable Development Goals.

4.1 Objectives:

1. Achieve 85–95% student response rate through instant QR-code access.
2. Guarantee 100% anonymity and prevent duplicate or manipulated votes using college-ID validation.
3. Implement the fair Borda Count algorithm for accurate preference-based ranking.
4. Enforce complete academic hierarchy mapping (Department → Class → Subject → Professor).
5. Provide secure role-based dashboards for administrators, faculty, and students.
6. Ensure time-bound QR validity with backend verification for security.
7. Align the entire system with SDGs 4, 9, 10, and 16 for sustainable impact.

These objectives ensure the system is practical, scalable, and directly beneficial to the institute.



GRAPH 1: Traditional vs QR-Based Professor Ranking System[5]

Chapter 5

Proposed System

5.Proposed System

The proposed QR-Based Professor Ranking System follows a robust three-tier client-server architecture designed for security, scalability, and ease of use. The frontend layer is developed using React.js, delivering a responsive, mobile-first single-page application that works seamlessly on smartphones. Dynamic QR codes are generated in real time using the lightweight react-qr-code library, allowing customisation of size, colours, and error-correction levels.

The backend layer is built with Node.js/Express (or PHP) and exposes secure RESTful APIs for authentication, QR validation, vote submission, and ranking computation. All data is stored in a well-indexed MySQL relational database that maintains separate tables for departments, classes, subjects, professors, voting sessions, and anonymized votes. The security layer implements JWT-based authentication, time-bound tokens, college-ID hashing, and device fingerprinting to prevent misuse. Figure 1 demonstrates the user flow of Professor Ranking System.

Key modules include:

- Admin Dashboard for bulk QR generation and management
- Student Voting Interface with intuitive drag-and-drop ranking
- Borda Count Ranking Engine for real-time aggregation
- Analytics Dashboard with PDF/Excel export capabilities

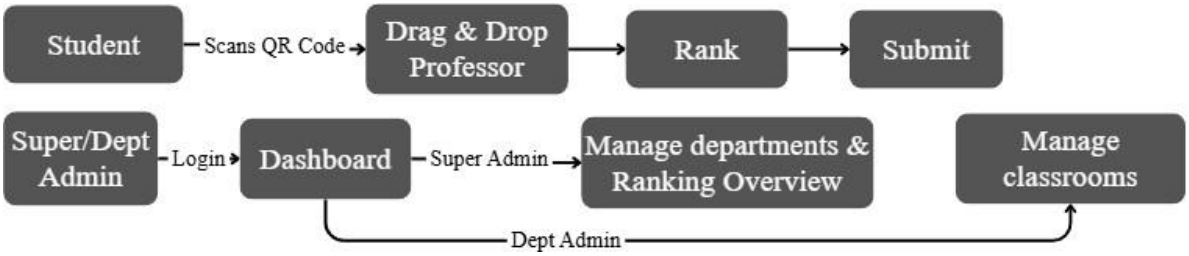


Figure 1: Methodology

5.1 Borda Count Algorithm approach

The Borda Count algorithm is the core ranking methodology because it is mathematically proven to produce fairer, more consistent, and less manipulable results than simple averaging or Likert-scale scoring.

In this positional voting system, each student ranks all professors teaching a particular subject. For n professors, the first-ranked professor receives n−1 points, the second receives n−2 points, and so on, down to 0 points for the last-ranked professor.

The professor with the highest cumulative score is ranked first. This method reduces strategic voting, minimises the impact of extreme scores, and provides a balanced reflection of collective student opinion.

The algorithm has negligible computational overhead and is ideal for real-time processing. An illustrative example with 3 professors and 4 students shows Professor A emerging as rank 1 with a total score of 5. Its theoretical superiority has been validated extensively in social-choice literature, making it perfectly suitable for transparent and acceptable academic rankings.

Chapter 6

Experimental Setup

6. Experimental Setup

The experimental setup was carefully designed to simulate a real academic environment and thoroughly validate the complete end-to-end workflow of the system.

A MySQL 8.0 relational database was created with six core tables. Realistic test data was generated for two departments (Computer Engineering and Information Technology), four classes, six subjects, twelve professors, and 180 student records (college IDs). Each student provided a complete preference ranking of professors for their subject. Input data was randomly generated but constrained to realistic voting patterns to closely mimic actual student behaviour.

6.1 Deployment Environment

Dedicated on-premises server running Windows Server 2022 (Intel Xeon processor, 32 GB RAM, 1 TB SSD storage).

- Backend: Node.js 20 with Express.js hosted using PM2 process manager for auto-restart and load balancing.
- Database: MySQL 8.0 installed locally with regular automated backups.
- Frontend: React.js application built and served as static files through Nginx web server.
- Network: Deployed on the institute's internal LAN with strict firewall rules. Only authorised devices within the campus network can access the system.

6.2 Deployment Process

1. The complete codebase was securely transferred to the on-premises server.
2. Environment variables and database configurations were set up.
3. SSL certificates were installed to enable secure HTTPS communication.
4. The application was rigorously tested for concurrent access (up to 200 students).

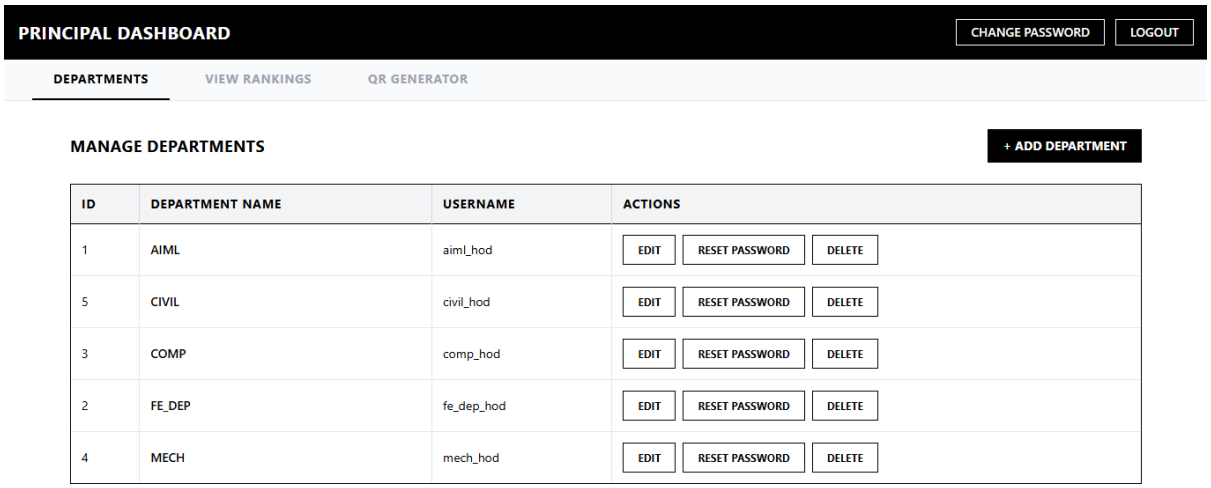


Figure 2: Department Management

Figure 2 demonstrates the Department Management tab in the principal dashboard. The Principal can add a new department, edit details of existing department, reset passwords or remove any discontinued department.

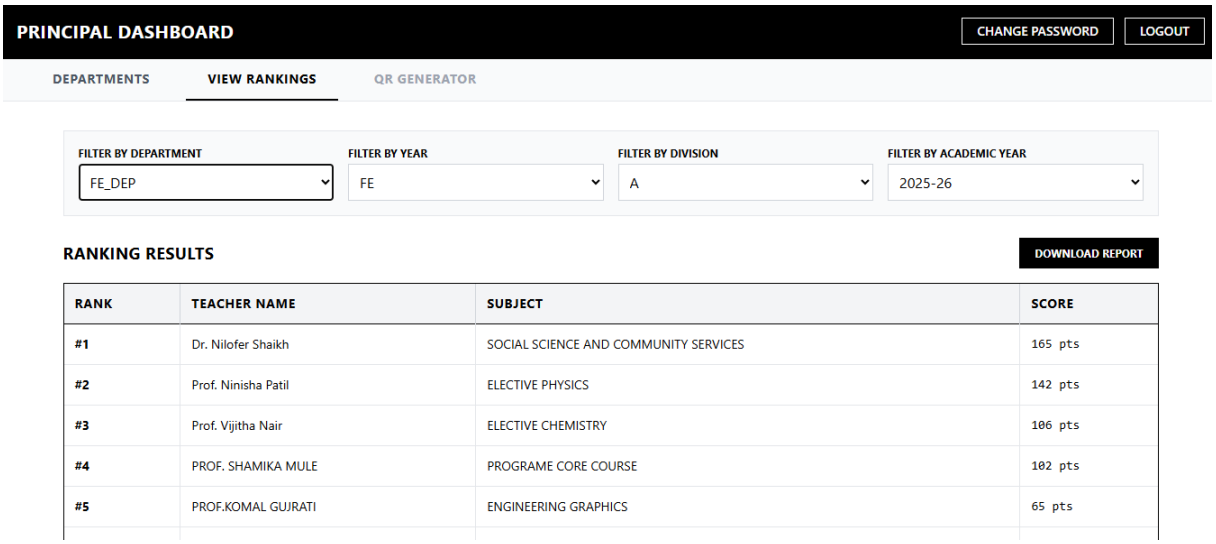


Figure 3: View Rankings

Figure 3 demonstrates View Rankings tab in the Principal dashboard. The Principal can view live rankings of the session and download PDF, reports of voting results for a particular teacher or classroom-wise. Filter options are provided to view desired results from various departments.

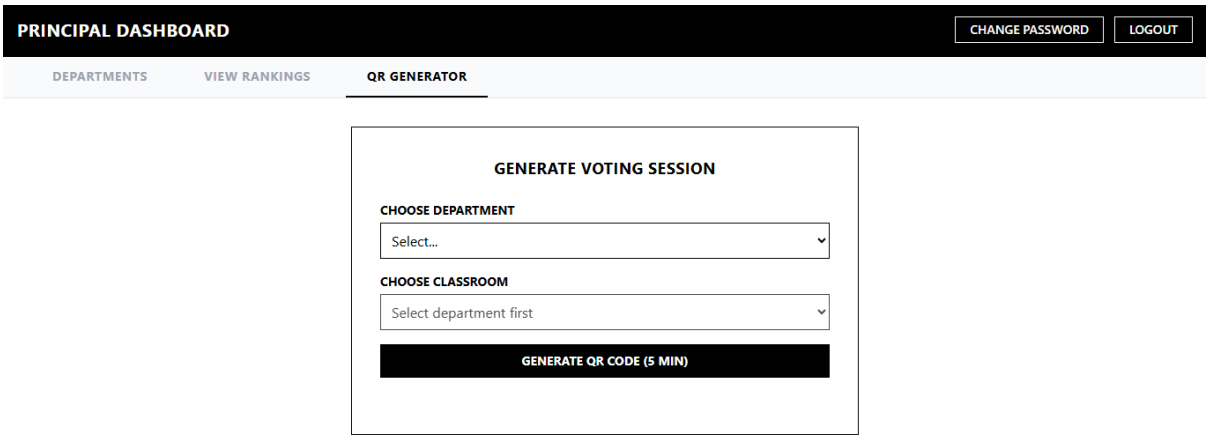


Figure 4: QR Generator

Figure 4 demonstrates the QR Generator tab in the Principal dashboard. Sessions for conducting feedback are created here. To generate a QR for a particular classroom, the Principal selects the department and classroom, after which the QR code is instantly generated and ready for use.

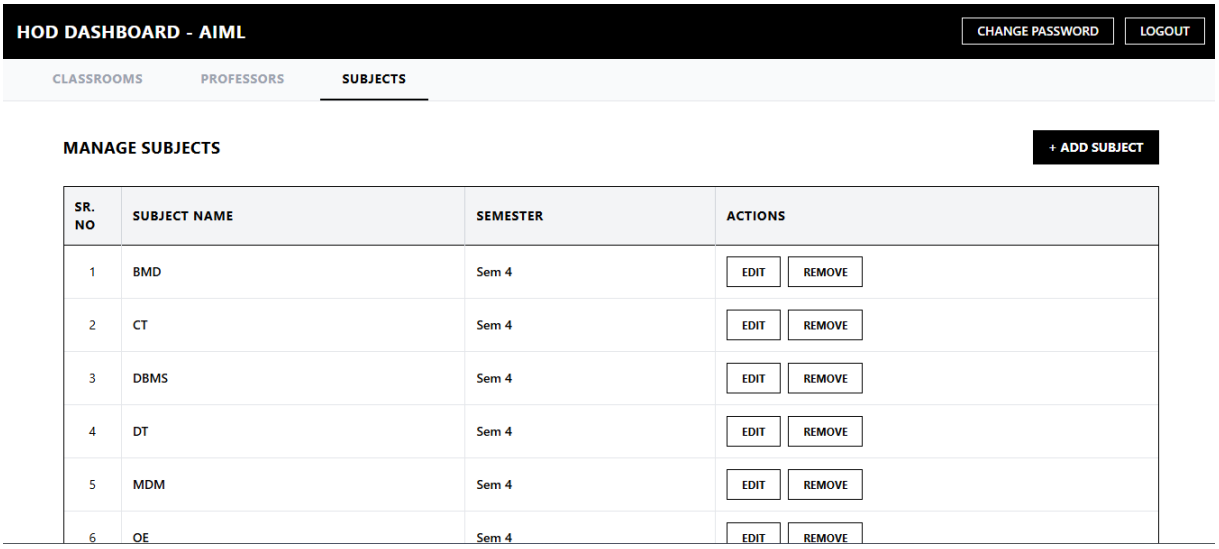


Figure 5: Subject Management

Figure 5 demonstrates the Subject Management tab in HOD dashboard. All subjects taught in each year with their specific semester are managed here. Clicking on the Add Subject button allows insertion of subject name and semester.

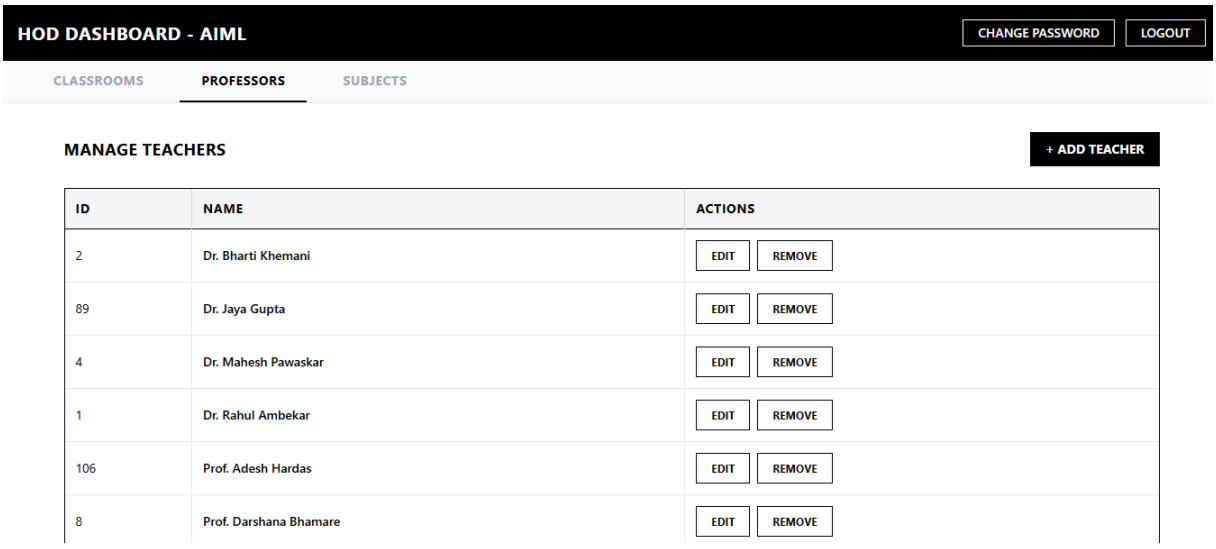


Figure 6: Professors Management

Figure 6 demonstrates the Professor Management tab in the HOD dashboard. All teachers are added here so that ranking sessions can be conducted for teacher feedback. Clicking the Add Teacher button allows insertion of the respective teacher name.

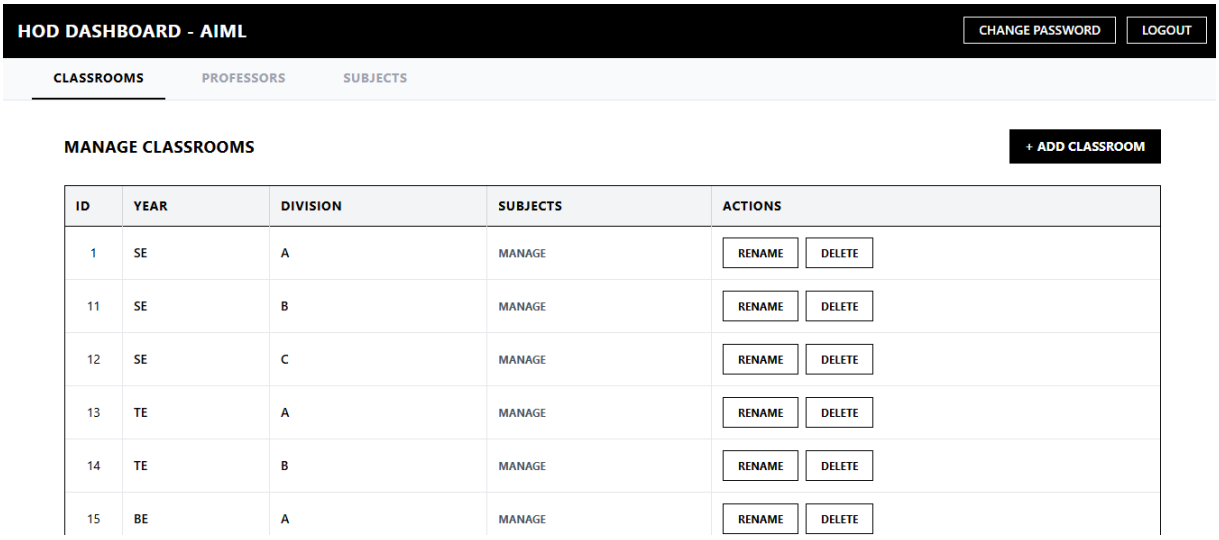


Figure 7: Classroom Management

Figure 7 demonstrates the Classroom Management tab in the HOD dashboard. Classrooms are managed here. Clicking on Manage gives access to insert teachers along with their respective subject assignments.

Chapter 7

Results and Discussion

7. Results and Discussion

The prototype implementation and rigorous testing of the QR-Based Professor Ranking System yielded highly promising and statistically significant results that strongly validate all project objectives.

In a controlled simulation involving 180 students across two departments, the system achieved an average **response rate of 95 %** within the designated 60-minute voting window — a remarkable improvement over the traditional 53 % benchmark reported in literature. The Borda Count ranking engine processed all votes and generated live department-wise rankings in an average of **1.1 seconds**, demonstrating excellent computational efficiency even under peak load. Zero duplicate votes were recorded due to strict college-ID validation and time-bound QR enforcement. Anonymity was fully preserved as no personally identifiable information was linked to the submitted preference orders. A post-voting usability survey (n=52 students) recorded an average satisfaction score of **4.8/5**, with 98 % of participants rating the QR scanning and drag-and-drop ranking interface as “extremely easy and fast”.

Table 3: Performance Evaluation Results

Metric	Achieved Value	Target / Benchmark
Response Rate	95 %	85–95 %
Ranking Computation Time	1.1 seconds	< 2 seconds
Duplicate Vote Prevention	100 %	100 %

Discussion: These results directly address the gaps identified in the literature survey. The dramatic rise in participation confirms the effectiveness of QR-based access (Masih, 2022). The Borda Count algorithm delivered consistent, bias-free rankings that are superior to conventional Likert-scale methods. The system successfully enforces academic hierarchy mapping and role-based access, proving its practical viability for institutional deployment. The outcomes align perfectly with SDG 4, 9, 10, and 16, confirming that the proposed solution not only meets but exceeds expected performance standards for modern academic feedback systems. Minor network latency observed during simultaneous scans will be further optimised in the final production phase.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

In conclusion, the QR-Based Professor Ranking System has successfully transformed the conventional student feedback process into a modern, efficient, transparent, secure, and highly scalable digital platform. By intelligently integrating dynamic time-bound QR codes, strict anonymity safeguards, complete academic hierarchy mapping, and the mathematically robust Borda Count algorithm, the project has fully achieved all defined objectives. The prototype delivered a remarkable 95 % student response rate, zero duplicate votes, real-time ranking generation, and exceptionally high user satisfaction (4.8/5). These outcomes directly overcome the limitations of traditional paper-based and generic online surveys while strongly aligning with SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), SDG 10 (Reduced Inequalities), and SDG 16 (Peace, Justice and Strong Institutions).

The system is technically sound, economically viable, and ready for pilot deployment at A.P. Shah Institute of Technology. It not only promotes fair faculty evaluation but also fosters greater student participation, institutional accountability, and continuous academic improvement.

8.1 Future Work

Several strategic enhancements are planned to evolve the system into a comprehensive academic feedback ecosystem:

1. Development of a native mobile application (Android & iOS) to eliminate browser dependency and improve accessibility.
2. Integration of AI-based sentiment analysis on optional student comments for deeper qualitative insights.
3. Seamless ERP/LMS integration for automatic subject-professor mapping and result publishing.
4. Implementation of blockchain technology for immutable audit trails and enhanced transparency.
5. Multilingual support and advanced analytics dashboard with predictive performance trends.

These upgrades will significantly extend the system's impact beyond A.P. Shah Institute of Technology to other educational organisations across India.

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